

# LABORATORY OBSERVATION

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Branch:** CSM

**Section:** B

**Task No.:** 2

**Student Name:** K. Saranya

**Roll No.:** A24126552090

---

## 1. TITLE

Build an Interactive Student Profile Manager Using JavaScript and DOM Manipulation

## 2. AIM

To build a webpage that collects student details through input fields and, on a button click, dynamically generates a styled student profile card using JavaScript DOM manipulation, without reloading the page.

## 3. OBJECTIVE

The objectives of this experiment are:

- To understand how to structure input fields and a button using HTML5 elements.
- To use a JavaScript class to model a real-world entity such as a Student.
- To select DOM elements using `document.getElementById()`.
- To handle a button click using `addEventListener("click", ...)`.
- To read and clean up values typed into input fields using `.value` and `.trim()`.
- To validate that all fields are filled before creating a profile, using a conditional check and `alert()`.
- To dynamically create and style elements using `document.createElement()`.
- To append dynamically created elements to the page using `appendChild()`.
- To reuse a helper function (`createDetail`) to avoid repetitive code when building each detail row.
- To apply CSS3 styling — gradients, box-shadow, hover/focus effects and a keyframe animation — for a polished, card-based interface.

## 4. THEORY

A JavaScript class is a blueprint for creating objects that share the same shape. The constructor runs once when a new object is created with the `new` keyword, and sets up its properties — here, a Student's name, rollNumber, department, and cgpa.

The DOM (Document Object Model) represents the page as a tree of elements that JavaScript can read and change. Instead of hardcoding the profile card in the HTML, this program builds it piece by piece at runtime, based on whatever the user typed into the input fields, and inserts it into an initially empty `<section id="profileContainer">`.

The key methods and concepts used in this program are:

- `addEventListener("click", fn)` runs a function every time the Create Student Profile button is clicked.
- `document.getElementById()` selects a specific element from the page by its id attribute.
- `.value.trim()` reads the current text in an input field and removes leading/trailing whitespace.
- A simple if condition checks whether any of the four trimmed values is an empty string; if so, an `alert()` is shown and the function returns early.
- `document.createElement()` and `appendChild()` create a new, empty element in memory and then attach it into the visible page.
- A reusable `createDetail(label, value, extraClass)` function builds one label/value row at a time, avoiding repeated DOM-creation code.
- `container.innerHTML = ""` clears any previously generated profile card before a new one is added.
- CSS `:focus` and `:hover` style an input or button only while it is actively selected or pointed at, and a `slideDown` keyframe animation fades and slides the new profile card into view.

The basic flow of the page is:

*User fills the input fields → Create Student Profile button is clicked → input values are read and trimmed → empty fields are checked → a Student object is created → a profile card is built with DOM methods → the card is appended to the page.*

## 5. ALGORITHM / PROCEDURE

1. Create three files: `index.html`, `style.css` and `script.js`, and link `style.css` in the head and `script.js` at the end of the body.
2. Add a form-style card with labelled inputs for Student Name, Roll Number, Department and CGPA, plus a Create Student Profile button.
3. Add an empty `<section id="profileContainer">` where the generated profile card will be inserted.
4. In `script.js`, define a `Student` class with a constructor that stores `name`, `rollNumber`, `department` and `cgpa`.
5. Select the input fields, the button and the container element using `document.getElementById()`.
6. Add a click event listener on the Create Student Profile button.
7. Inside the handler, read the current values from each input field using `.value.trim()`.
8. Check whether any field is empty; if so, show an alert and stop (return) without creating a profile.
9. Create a new `Student` object using the values just read.
10. Clear any existing content in the container using `profileContainer.innerHTML = ""`.
11. Create a profile card `<div>`, and inside it a header (icon + heading) and a details section.
12. Write a `createDetail()` helper that builds one label/value row as a `<div>` and returns it.
13. Call `createDetail()` once for each of Name, Roll No, Department and CGPA, and append each row to the details section.
14. Append the header and details section to the card, then append the card to the container so it appears on the page.
15. Save the files and open `index.html` in a browser to test with different input values.

## 6. PROGRAM

### index.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <!-- Makes the webpage responsive on different screen sizes -->
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Student Profile Manager</title>
8      <!-- Linking the external CSS file -->
9      <link rel="stylesheet" href="style.css">
10 </head>
11 <body>
12     <!-- Main container for the Student Profile Manager -->
13     <div class="container">
14         <!-- Page heading -->
15         <header>
16             <h1>🎓 Student Profile Manager</h1>
17             <p>Create and view a student profile dynamically using JavaScript</p>
18         </header>
19         <!-- Form section for entering student details -->
20         <section class="form-card">
21             <h2>Enter Student Details</h2>
22             <div class="input-group">
23                 <label for="name">Student Name</label>
24                 <input type="text" id="name" placeholder="Enter student name">
25             </div>
26             <div class="input-group">
27                 <label for="roll">Roll Number</label>
28                 <input type="text" id="roll" placeholder="Enter roll number">
29             </div>
30             <div class="input-group">
31                 <label for="department">Department</label>
32                 <input type="text" id="department" placeholder="Enter department">
33             </div>
34             <div class="input-group">
35                 <label for="cgpa">CGPA</label>
36                 <input type="number" id="cgpa" step="0.01" min="0" max="10"
37                     placeholder="Enter CGPA">
38             </div>
39             <!-- Button used to create the profile dynamically -->
40             <button id="createBtn">Create Student Profile</button>
41         </section>
42         <!--
43             This empty section acts as a container.
44             JavaScript will dynamically create the student profile
45             and insert it into this section.
46         -->
47         <section id="profileContainer"></section>
48     </div>
49     <!-- Linking the external JavaScript file -->
50     <script src="script.js"></script>
51 </body>
52 </html>
```

### style.css

```
1  /* Remove default spacing and apply a common font */
2  * {
3      margin: 0;
4      padding: 0;
5      box-sizing: border-box;
6  }
7  /* Styling the complete webpage */
8  body {
9      font-family: Arial, sans-serif;
```

```
10     min-height: 100vh;
11     background: linear-gradient(135deg, #667eea, #764ba2);
12     display: flex;
13     justify-content: center;
14     align-items: center;
15     padding: 30px;
16 }
17 /* Main application container */
18 .container {
19     width: 100%;
20     max-width: 850px;
21 }
22 /* Page heading */
23 header {
24     text-align: center;
25     color: white;
26     margin-bottom: 25px;
27 }
28 header h1 {
29     font-size: 32px;
30     margin-bottom: 8px;
31 }
32 header p {
33     font-size: 15px;
34     opacity: 0.9;
35 }
36 /* Card containing the input form */
37 .form-card {
38     background: white;
39     padding: 30px;
40     border-radius: 18px;
41     box-shadow: 0 10px 30px rgba(0, 0, 0, 0.2);
42     margin-bottom: 25px;
43 }
44 .form-card h2 {
45     text-align: center;
46     margin-bottom: 22px;
47 }
48 /* Each input field */
49 .input-group {
50     margin-bottom: 16px;
51 }
52 .input-group label {
53     display: block;
54     font-weight: bold;
55     margin-bottom: 6px;
56 }
57 .input-group input {
58     width: 100%;
59     padding: 12px;
60     border: 1px solid #ccc;
61     border-radius: 8px;
62     font-size: 15px;
63 }
64 /* Input focus effect */
65 .input-group input:focus {
66     outline: none;
67     border-color: #667eea;
68     box-shadow: 0 0 5px rgba(102, 126, 234, 0.4);
69 }
70 /* Create button */
71 button {
72     width: 100%;
73     padding: 13px;
74     border: none;
75     border-radius: 9px;
76     background: #667eea;
77     color: white;
78     font-size: 16px;
```

```

79     font-weight: bold;
80     cursor: pointer;
81     transition: 0.3s;
82 }
83 button:hover {
84     background: #4f63c7;
85     transform: translateY(-2px);
86 }
87 /* Dynamically generated student profile */
88 .profile-card {
89     background: white;
90     border-radius: 18px;
91     overflow: hidden;
92     box-shadow: 0 10px 30px rgba(0, 0, 0, 0.2);
93     animation: slideIn 0.5s ease;
94 }
95 /* Profile heading */
96 .profile-header {
97     background: linear-gradient(135deg, #667eea, #764ba2);
98     color: white;
99     text-align: center;
100    padding: 25px;
101 }
102 .profile-header .icon {
103     font-size: 45px;
104     margin-bottom: 8px;
105 }
106 .profile-header h2 {
107     font-size: 24px;
108 }
109 /* Student information section */
110 .profile-details {
111     padding: 25px;
112 }
113 /* Individual student information row */
114 .detail {
115     display: flex;
116     justify-content: space-between;
117     padding: 14px 5px;
118     border-bottom: 1px solid #eee;
119 }
120 .detail:last-child {
121     border-bottom: none;
122 }
123 .detail .label {
124     font-weight: bold;
125     color: #555;
126 }
127 .detail .value {
128     color: #333;
129     text-align: right;
130 }
131 /* CGPA highlight */
132 .cgpa {
133     font-weight: bold;
134     color: #667eea !important;
135 }
136 /* Animation for dynamically created profile */
137 @keyframes slideIn {
138     from {
139         opacity: 0;
140         transform: translateY(20px);
141     }
142     to {
143         opacity: 1;
144         transform: translateY(0);
145     }
146 }
147 /* Responsive design for smaller screens */

```

```

148 @media (max-width: 600px) {
149     body {
150         padding: 15px;
151     }
152     header h1 {
153         font-size: 25px;
154     }
155     .form-card {
156         padding: 20px;
157     }
158     .detail {
159         flex-direction: column;
160         gap: 5px;
161     }
162     .detail .value {
163         text-align: left;
164     }
165 }

```

## script.js

```

1  // Creating a Student class
2  class Student {
3      // Constructor initializes the properties of a student
4      constructor(name, rollNumber, department, cgpa) {
5          this.name = name;
6          this.rollNumber = rollNumber;
7          this.department = department;
8          this.cgpa = cgpa;
9      }
10 }
11 // Selecting the required HTML elements using their IDs
12 const nameInput = document.getElementById("name");
13 const rollInput = document.getElementById("roll");
14 const departmentInput = document.getElementById("department");
15 const cgpaInput = document.getElementById("cgpa");
16 const createButton = document.getElementById("createBtn");
17 const profileContainer = document.getElementById("profileContainer");
18 // Adding a click event to the Create Profile button
19 createButton.addEventListener("click", function () {
20
21     const name = nameInput.value.trim();
22     const rollNumber = rollInput.value.trim();
23     const department = departmentInput.value.trim();
24     const cgpa = cgpaInput.value.trim();
25     // Check whether all fields are filled
26     if (name === "" || rollNumber === "" ||
27         department === "" || cgpa === "") {
28         alert("Please enter all student details.");
29         return;
30     }
31
32     const student = new Student(
33         name,
34         rollNumber,
35         department,
36         cgpa
37     );
38
39     profileContainer.innerHTML = "";
40
41     const profileCard = document.createElement("div");
42     profileCard.className = "profile-card";
43     // Creating the profile heading section
44     const profileHeader = document.createElement("div");
45     profileHeader.className = "profile-header";
46     const icon = document.createElement("div");
47     icon.className = "icon";
48     icon.textContent = "👤";

```

```

49     const heading = document.createElement("h2");
50     heading.textContent = "Student Profile";
51     // Adding heading elements to the profile header
52     profileHeader.appendChild(icon);
53     profileHeader.appendChild(heading);
54     // Creating the student details section
55     const profileDetails = document.createElement("div");
56     profileDetails.className = "profile-details";
57
58     function createDetail(labelText, valueText, extraClass = "") {
59         const detail = document.createElement("div");
60         detail.className = "detail";
61         const label = document.createElement("span");
62         label.className = "label";
63         label.textContent = labelText;
64         const value = document.createElement("span");
65         value.className = "value " + extraClass;
66         value.textContent = valueText;
67         detail.appendChild(label);
68         detail.appendChild(value);
69         return detail;
70     }
71
72     profileDetails.appendChild(
73         createDetail("Name", student.name)
74     );
75     profileDetails.appendChild(
76         createDetail("Roll No", student.rollNumber)
77     );
78     profileDetails.appendChild(
79         createDetail("Department", student.department)
80     );
81     profileDetails.appendChild(
82         createDetail("CGPA", student.cgpa, "cgpa")
83     );
84
85     profileCard.appendChild(profileHeader);
86     profileCard.appendChild(profileDetails);
87     profileContainer.appendChild(profileCard);
88 });

```

## 7. CODE EXPLANATION

| Code                                    | Explanation                                                                                                |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------|
| class Student { constructor(...) }      | Defines a blueprint for creating student objects with name, rollNumber, department, and cgpa.              |
| new Student(...)                        | Creates a new object instance using the values read from the input fields.                                 |
| document.getElementById()               | Selects a specific element from the page by its id attribute.                                              |
| addEventListener("click", fn)           | Runs the given function every time the Create Student Profile button is clicked.                           |
| .value.trim()                           | Reads the current text typed into an input field and strips extra whitespace.                              |
| if (... === "") { alert(...); return; } | Validates that all four fields are filled before a profile is created; stops the function early otherwise. |
| profileContainer.innerHTML = ""         | Clears any previously generated profile card before adding a new one.                                      |
| document.createElement()                | Creates a new, empty HTML element in memory.                                                               |
| element.textContent                     | Sets the visible text inside an element.                                                                   |
| appendChild()                           | Attaches a created element as a child of another element in the DOM.                                       |
| createDetail(label, value, extraClass)  | Reusable helper that builds one label/value detail row, avoiding repeated code for each field.             |
| :focus / :hover                         | Styles an input while it is being typed into, or the button while the pointer is over it.                  |

| Code               | Explanation                                                                      |
|--------------------|----------------------------------------------------------------------------------|
| @keyframes slideIn | Animates the profile card with a fade-and-slide-up effect when it first appears. |

## 8. TEST CASES AND EXECUTION DATA

The page was opened in a browser and tested with different inputs; the results were recorded below.

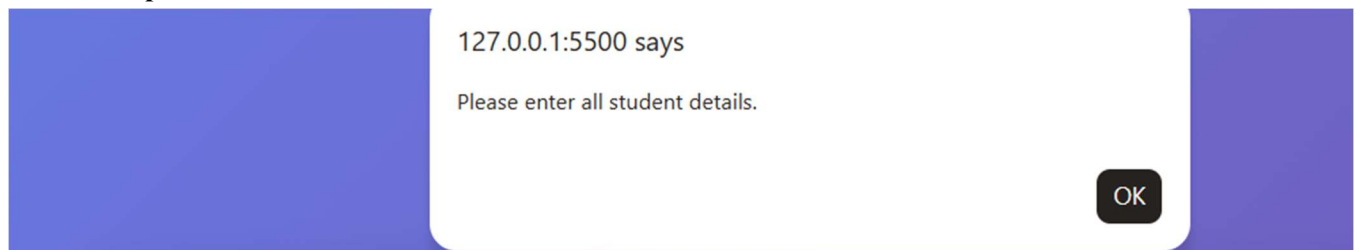
Additional Validation Test

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| <b>Test Case:</b> TC-01                                                                                  |
| <b>Action:</b> Open index.html in a web browser.                                                         |
| <b>Expected Result:</b> Empty form displayed with 4 labelled fields and a Create Student Profile button. |
| <b>Actual Result:</b> Form displayed correctly with empty input fields.                                  |
| <b>Status:</b> Pass                                                                                      |

**TC-01 Output:**

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| <b>Test Case:</b> TC-02                                                                         |
| <b>Action:</b> Click Create Student Profile with all fields empty.                              |
| <b>Expected Result:</b> An alert asks to enter all student details; no profile card is created. |
| <b>Actual Result:</b> Alert shown as expected; script execution halts early.                    |
| <b>Status:</b> Pass                                                                             |

**TC-02 Output:**



|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <b>Test Case:</b> TC-03                                                                                      |
| <b>Action:</b> Fill all fields and click Create Student Profile.                                             |
| <b>Expected Result:</b> A profile card appears below the form with the entered values; page does not reload. |
| <b>Actual Result:</b> Profile card displayed correctly with smooth slideIn animation.                        |
| <b>Status:</b> Pass                                                                                          |

**TC-03 Output:**



## Student Profile

|            |               |
|------------|---------------|
| Name       | Kundi Saranya |
| Roll No    | A24126550290  |
| Department | CSE(AI&ML)    |
| CGPA       | 9.27          |

|                                                                                         |
|-----------------------------------------------------------------------------------------|
| <b>Test Case:</b> TC-04                                                                 |
| <b>Action:</b> Click Create Student Profile a second time with different values.        |
| <b>Expected Result:</b> Old profile card is removed and replaced with the new one.      |
| <b>Actual Result:</b> Card replaced correctly without appending multiple profile cards. |
| <b>Status:</b> Pass                                                                     |

#### TC-04 Output:



## Student Profile

|            |              |
|------------|--------------|
| Name       | Rahul V      |
| Roll No    | A24126550291 |
| Department | ECE          |
| CGPA       | 8.95         |

|                                                                                |
|--------------------------------------------------------------------------------|
| <b>Test Case:</b> TC-05                                                        |
| <b>Action:</b> Click into any input field.                                     |
| <b>Expected Result:</b> Border color changes and a soft glow appears on focus. |
| <b>Actual Result:</b> Focus effect displayed correctly.                        |
| <b>Status:</b> Pass                                                            |

#### TC-05 Output:

**Student Name**

|                                                                           |
|---------------------------------------------------------------------------|
| <b>Test Case:</b> TC-06                                                   |
| <b>Action:</b> Hover over the Create Student Profile button.              |
| <b>Expected Result:</b> Background darkens and the button lifts slightly. |

**Actual Result:** Hover transition executed smoothly.

**Status:** Pass

**TC-06 Output:**

Before hover effect

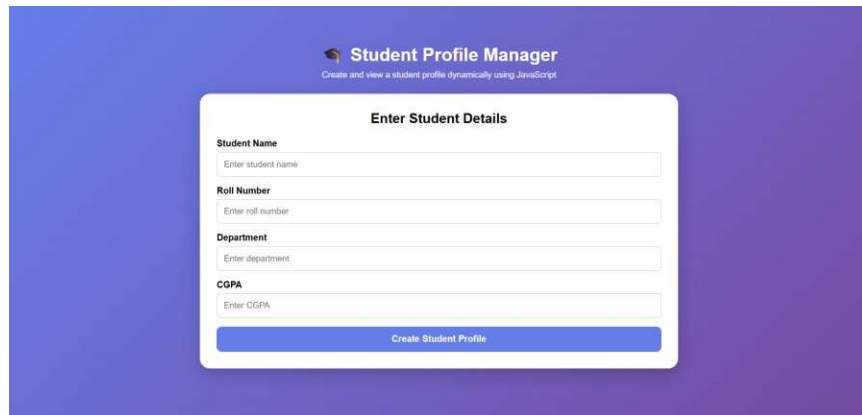
Create Student Profile

After hover effect

Create Student Profile

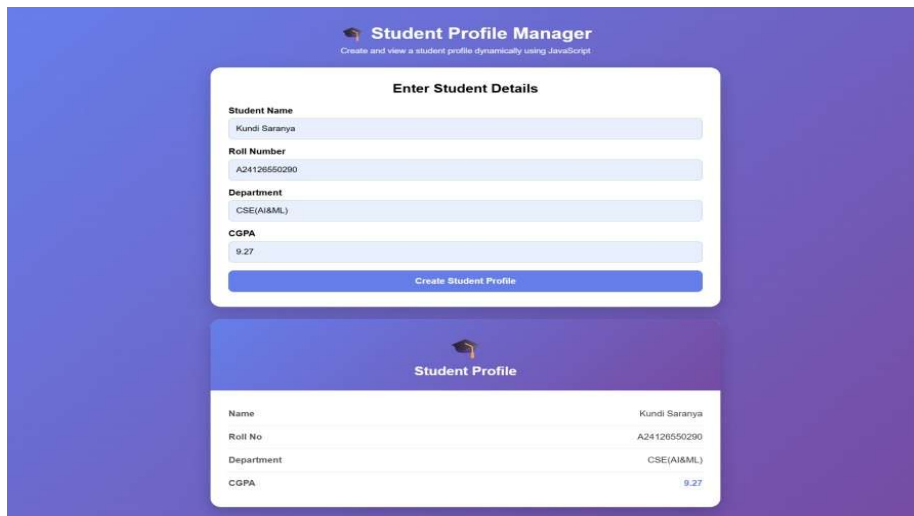
## 9. OUTPUT

### Output 1: Empty Student Details Form



The screenshot shows a web application titled "Student Profile Manager" with the subtitle "Create and view a student profile dynamically using JavaScript". The main form is titled "Enter Student Details" and contains four input fields: "Student Name", "Roll Number", "Department", and "CGPA". Each field has a placeholder text "Enter [field name]". At the bottom of the form is a blue button labeled "Create Student Profile".

### Output 2: Filled Form and Generated Profile Card (after clicking Create Student Profile)



The screenshot shows the same web application as before, but now the form fields are filled with the following data: Student Name: Kundi Saranya, Roll Number: A24126550290, Department: CSE(AI&ML), and CGPA: 9.27. The "Create Student Profile" button is still present. Below the form, a new "Student Profile" card is displayed. The card has a blue header with a graduation cap icon and the title "Student Profile". It contains the same data as the form: Name: Kundi Saranya, Roll No: A24126550290, Department: CSE(AI&ML), and CGPA: 9.27.

## 10.RESULT

Thus, an interactive Student Profile Manager was successfully built using HTML, CSS and JavaScript DOM manipulation. Clicking the Create Student Profile button with all fields filled correctly created a Student object and rendered a styled profile card, complete with a gradient header, icon, and highlighted CGPA value, without reloading the page. Leaving any field empty correctly triggered a validation alert

instead of creating a profile.